

DeepFace Studio

AI face swap that places a visitor into a real campus photograph

A visitor enters their name, uploads or captures a photo, and picks a campus location. Their face is swapped into the curated photograph for that location, then returned as a preview plus a high-resolution download.

Tech stack

Frontend React 18 + Vite + TailwindCSS + Framer Motion (SPA, served by Flask)
Backend Python + Flask, background worker threads for the swap pipeline
Storage Supabase (location photographs) with a local Location/ folder fallback
Auth Firebase Google Sign-In — gates the Control Panel to an email allowlist

Models

buffalo_l (InsightFace) face detection + 5-point landmarks
inswapper_128 (InsightFace) the identity swap itself, internally 128x128
GFPGAN v1.4 restores facial detail lost to that 128px bottleneck
RealESRGAN x4plus super-resolution, applied to the head region
BiSeNet (facexlib) hair / skin / neck segmentation
HairFastGAN (hosted Space) optional hairstyle transfer — needs HF_TOKEN

Hosting

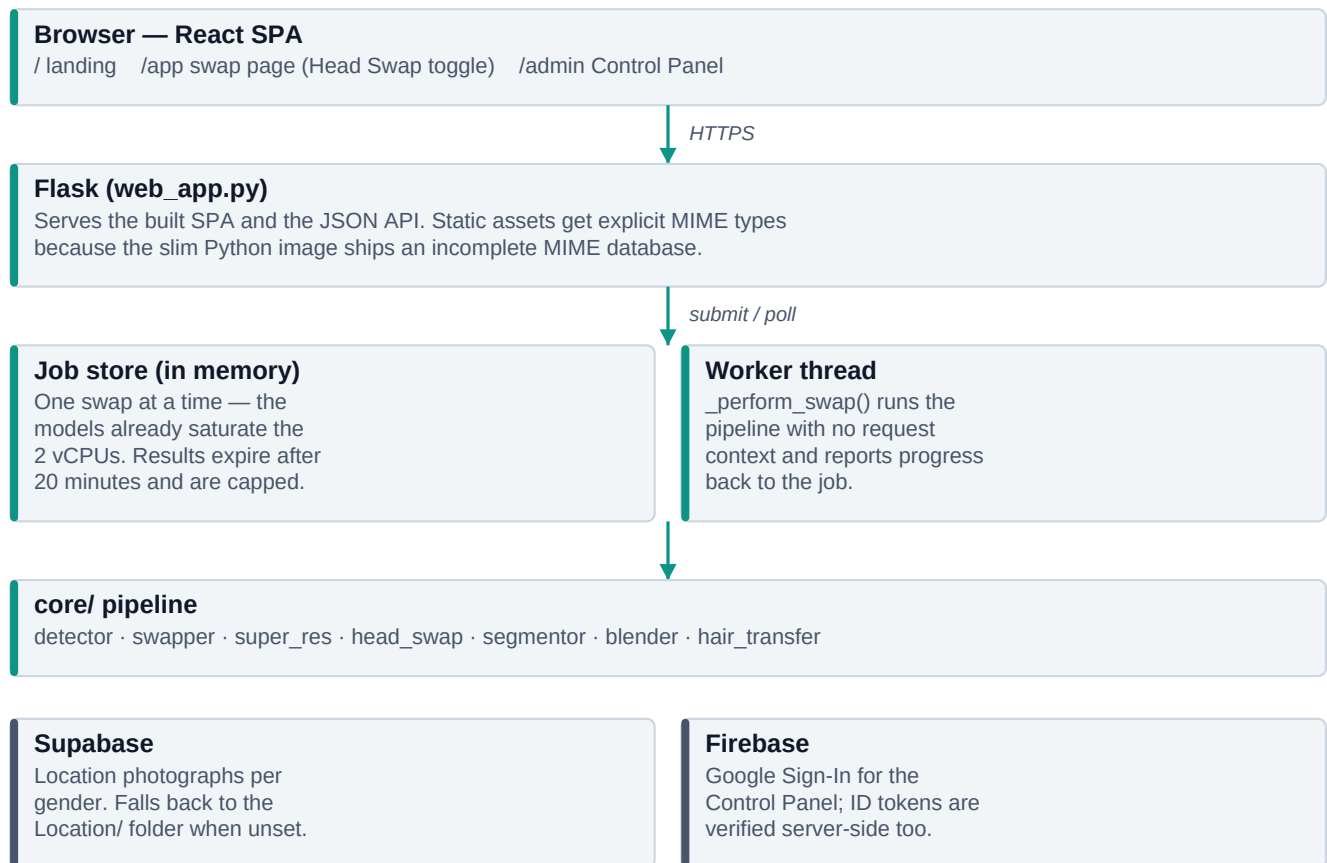
Hugging Face Space [aditya-raj19/FaceSWAP](https://aditya-raj19-faceswap.hf.space) (Docker SDK, port 7860)
Live <https://aditya-raj19-faceswap.hf.space>
Hardware cpu-basic (free tier); CUDA is used automatically when present
Deploy every push to main, via GitHub Actions

Why the pipeline is built the way it is

The swap model works at 128x128, so the head is the only part of the frame that loses real detail. Everything else is taken from the original photograph untouched, and super-resolution is spent on the head alone rather than the whole image.

System architecture

What talks to what, at request time



Nothing is written to disk during a swap — images are decoded, processed and returned in memory.

Request workflow

Why a swap is a job rather than a single request

The constraint

Full-quality output takes minutes on the free CPU tier — head super-resolution alone is about six. No HTTP request survives that, so the swap cannot be synchronous.

1 POST /api/swap

Validates and decodes the source, resolves the target from gender + location, creates a job and returns 202 with a job_id. Returns in about 1.5 seconds.



2 Worker thread picks up the job

Runs the pipeline, updating progress and a message at each stage.



3 GET /api/swap/status/<job_id>

Polled every 2 seconds. Returns state, progress and message while running.



4 state = done

The same payload the endpoint always returned: result_image, download_image, quality, delta_e, src_tone, tgt_tone, warnings, name.

In the UI

- The progress bar follows the server's real stage, not a timer animating a guess.
- The motivational captions keep rotating on their own — the wait is longer now.
- Failures surface the server's message instead of failing silently.

The swap pipeline

Stage by stage, with measured CPU cost on a 1086x1448 photograph

Working resolution is 1024 on CPU and 1536 on CUDA. Optional stages are marked.

STAGE	WHAT IT DOES	CPU
Prepare	Resize; pad a close-up until detectable; de-roll a tilted selfie so eyes are level.	~7s
Detect	buffalo_l on source and target; retries with CLAHE, then Haar.	~1s
Head swap (opt-in)	Transplants the whole head — shape and hair — instead of only the face.	varies
Face swap	inswapper_128 puts the source identity on the main target face.	~19s
Restore	GFPGAN rebuilds the detail the 128px swap threw away.	~55s
Blend seam	Laplacian pyramid over the face boundary. Must run before hair.	~14s
Hair (opt-in)	HairFastGAN only. No crude fallback — it smeared a blob over the head.	~30s
Glasses (opt-in)	Off by default: it disfigured sources who wear glasses.	~4s
Skin tone	Drives face, neck, arms and hands to the source complexion.	~3s
Score	Alignment, blend, delta-E, naturalness — measured before grain is added.	~2s
Harmonise	Adds the photo's grain over the parsed head so it isn't GAN-smooth.	~1s
Composite	Pastes only what changed back onto the untouched full-resolution photo.	~0.2s
Enhance	RealESRGAN on the head crop, Lanczos elsewhere, feathered between.	~370s

The two decisions that matter most

Composite, not upscale — the background and body are never touched by the swap, so they are taken from the original photograph. Background sharpness went from 34% of the original to 100%, and the delivered image is native resolution rather than a downscaled frame blown back up.

Deployment

From a push on main to a running Space

git push origin main

The default branch is the deploy trigger.



GitHub Actions

ci.yml pytest on a lightweight dependency subset, plus a frontend build
deploy.yml force-pushes the repository to the Hugging Face Space



Hugging Face Space

Rebuilds the Docker image on the new commit.



Docker build

Stage 1 node:20-slim builds the React app into static/react
Stage 2 python:3.10-slim installs CPU torch, then requirements-deploy.txt



startup.sh

Downloads model weights in the BACKGROUND and starts Flask immediately —
a container that doesn't answer on its port within ~60s is killed.



Flask on :7860

Debug off; models pre-warm on a daemon thread so the port binds at once.

Configuration held outside the repository

- HF_TOKEN — GitHub secret for deploying; also lifts the HairFastGAN GPU quota.
- SUPABASE_URL / SUPABASE_SERVICE_KEY / SUPABASE_BUCKET — location photographs.
- ALLOWED_ADMIN_EMAILS / PRIMARY_ADMIN_EMAILS — Control Panel access.
- FLASK_DEBUG — unset in production; set to true only for local development.

Verifying a deploy

Space state and commit: <https://huggingface.co/api/spaces/aditya-rAj19/FaceSWAP/runtime>
Expect "stage": "RUNNING" with sha matching the pushed commit, then probe / and /api/locations.

Behaviour and known limits

Things that are easy to misread from the code

CPU and GPU

Both paths run the same pipeline; only cost and working resolution differ.

CPU 1024px working resolution; RealESRGAN on the head crop (~6 min).

GPU 1536px; RealESRGAN available for the full frame; local HairFastGAN possible.

Full-frame super-resolution on CPU was measured at about 20 minutes, which is why it is confined to the head.

Hair transfer

Real hair transfer requires HairFastGAN. On the Space it connects to the public AIRI-Institute/HairFastGAN ZeroGPU Space anonymously and hits its quota, so it fails and the target's own hair is kept.

To enable it: set HF_TOKEN on the Space, or point HAIRFAST_SPACE at your own GPU Space. There is deliberately no fallback — warping hair in directly looked pasted.

Opt-in stages, and why

Head swap changes far more of the photo than a face swap; exposed as a toggle.

Hair only runs when HairFastGAN returns a portrait.

Glasses warped spectacles onto the face as a misaligned, translucent shape.

Each is a form field on /api/swap, so any of them can be turned back on per request.

Quality metrics

Reported per swap: alignment, blend, delta-E and naturalness. They are computed BEFORE the harmonise stage — that stage adds grain deliberately, and scoring after it penalised the pipeline for its own noise, understating every result.

Privacy

Uploads are decoded in memory and the result is returned as a data URI; nothing is written to disk. Finished jobs, which hold the images, expire after 20 minutes.